



Department of Computer Science and Engineering (Data Science)

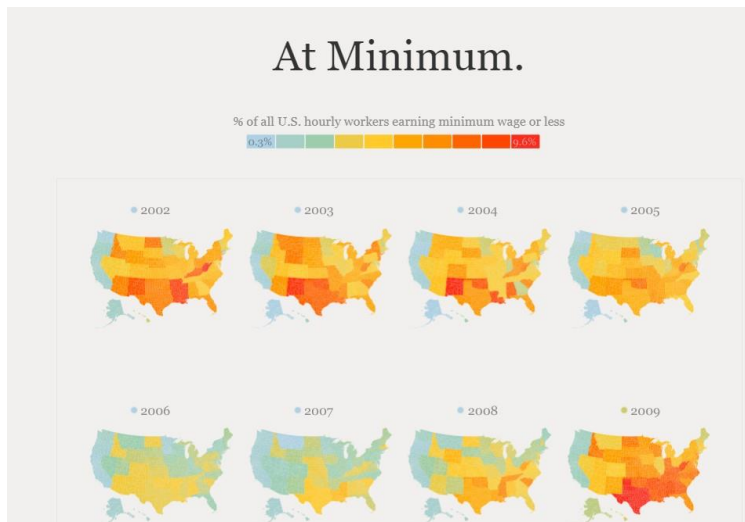
Experiment 2

(Data Visualization)

Aim: Apply maps, scatter plots on a given dataset and create a dashboard

Theory:

With technology advancements, content on maps and the maps themselves became digital, interactive, and more appealing as they're incorporated in data analysis and reporting. Seeing location data mapped and included in visualizations has both enhanced understanding by more audiences and offered a valuable, new context. Maps share geographic context that can be important to visualize results, something traditional bar charts and graphs fail to depict.



The scatter diagram graphs pairs of numerical data, with one variable on each axis, to look for a relationship between them. If the variables are correlated, the points will fall along a line or curve. The better the correlation, the tighter the points will hug the line.

Scatter Plots are often used to show additional numeric data in the form of Colors and Sizes, in which case they plot up to 4-dimensional data.



Shri Vile Parle Kelavani Mandal's

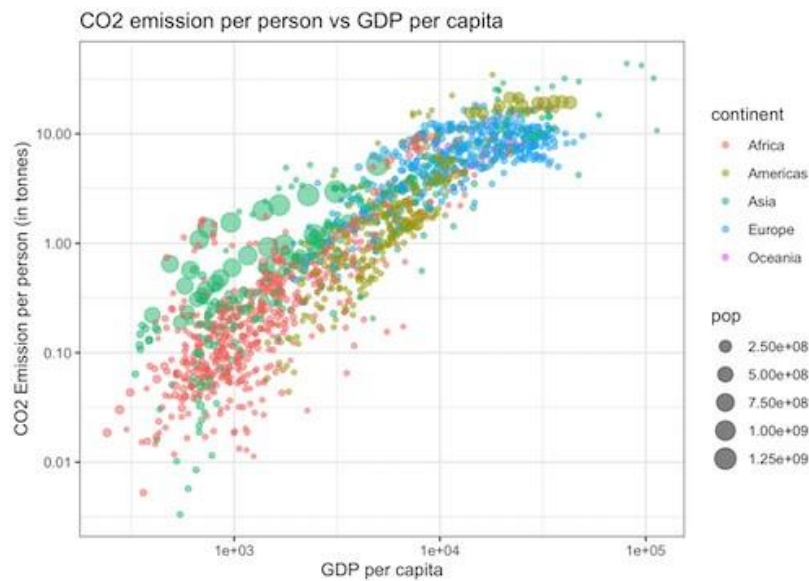
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



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Lab Assignments to complete in this session

Use the given dataset and perform the following tasks:

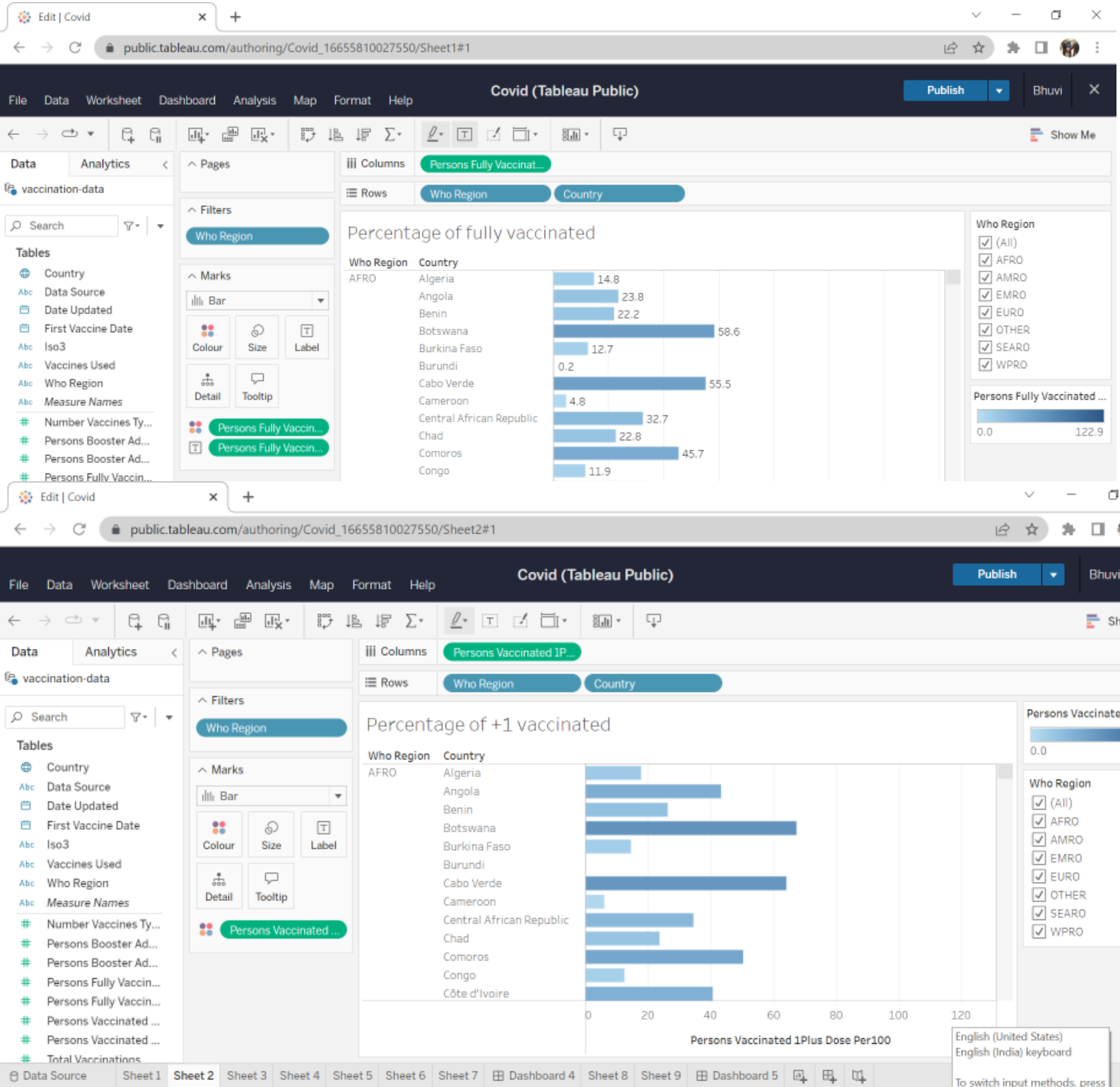
Dataset:

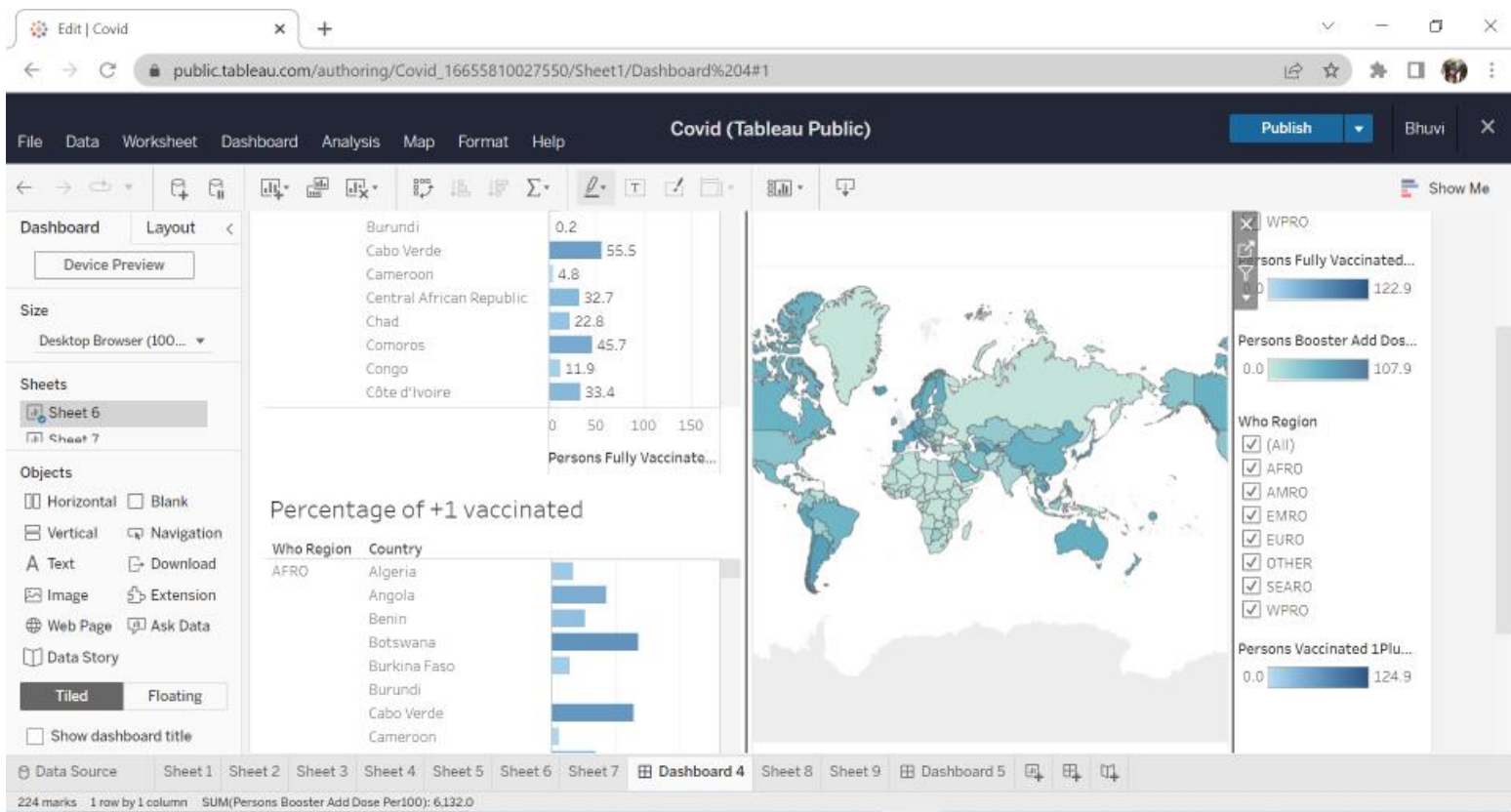
<https://covid19.who.int/who-data/vaccination-data.csv>

<https://basic-csv-wdc.herokuapp.com/>

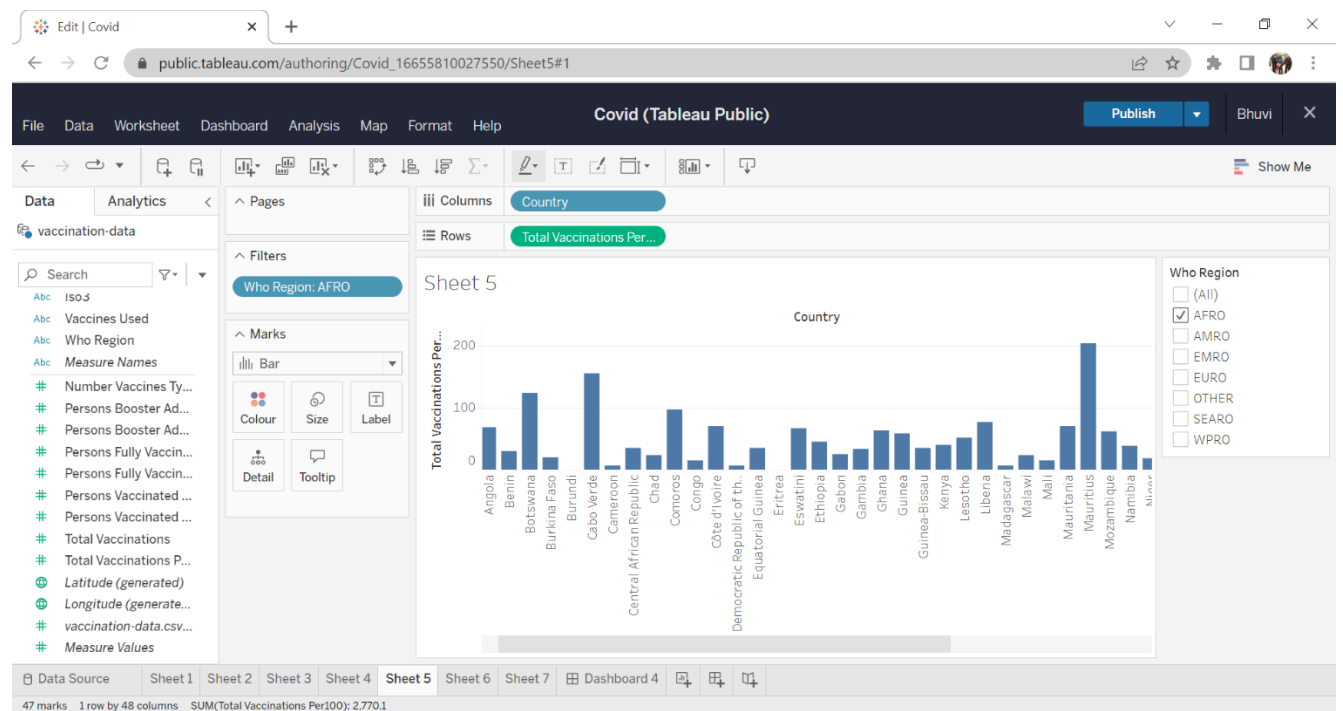
- The United Nations wants you to build a dashboard containing an overview of the vaccination progress, specifically 1+ doses and Fully vaccinated individuals, as a percentage of population vaccinated.
- The Covax vaccine scheme was used to supply vaccines to poorer countries like Africa. Has the scheme proven to be effective? - <https://www.bbc.com/news/56100076>
- Use Maps to visualize the number of people vaccinated per 100 population to visualize the countries and regions in dire need of vaccines. It should be noted that the data only covers WHO approved vaccines.
- Build a scatter plot to view the trend between 1+ Dose vaccinated and Fully vaccinated population, so the UN can advise countries that deviate from the recommended values to focus on getting a higher percentage of population fully vaccinated, or vice versa.
- The Dashboard should allow vaccine monitoring and filtering via WHO regions, allowing filtering for multiple regions.

Q1) a) Graphs based on the percentage of fully vaccinated & +1 vaccinated people in the countries of Africa. Dashboard connecting the two is also made in such a way that it gives all the necessary information after a particular country is clicked.

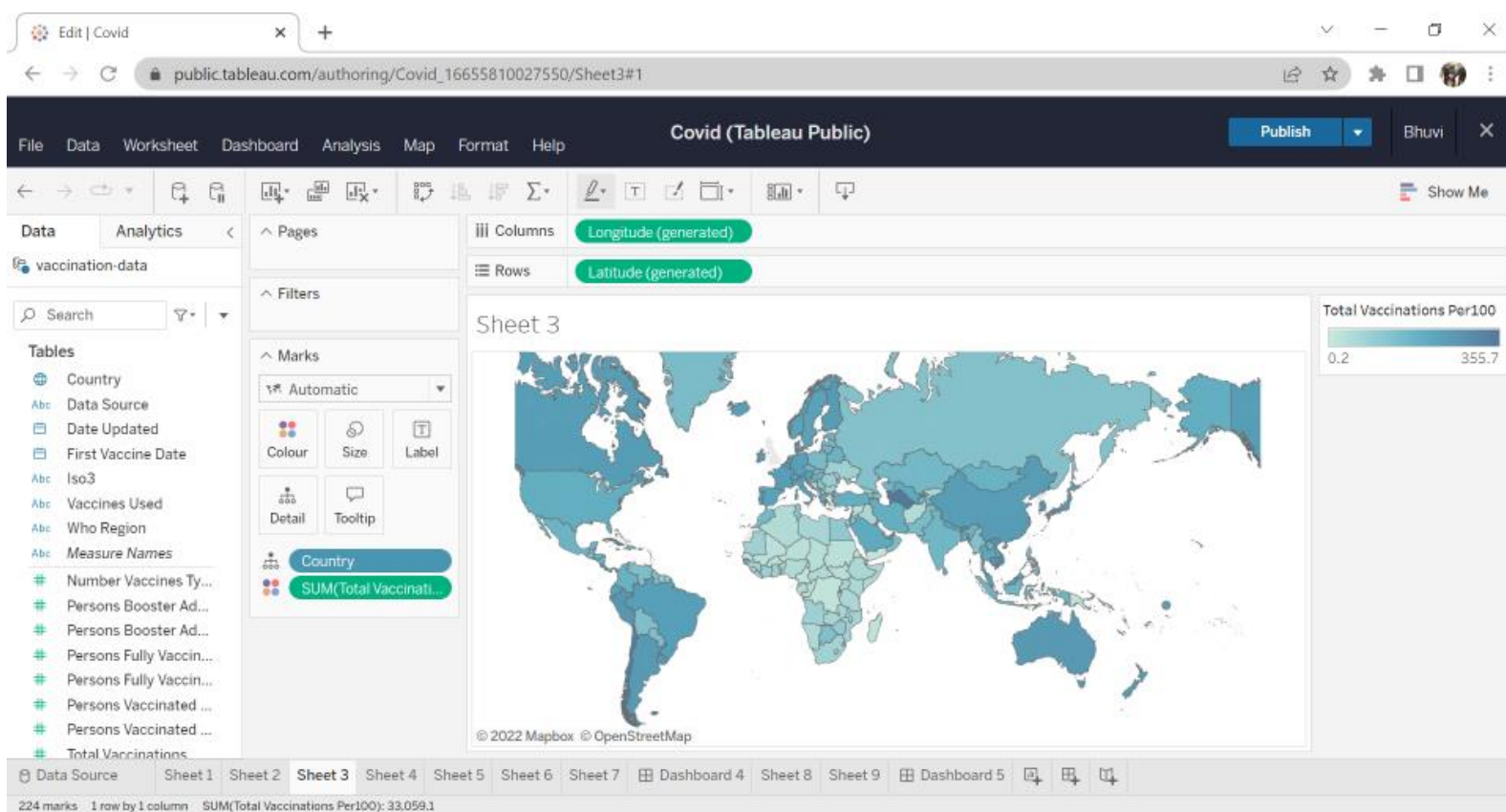




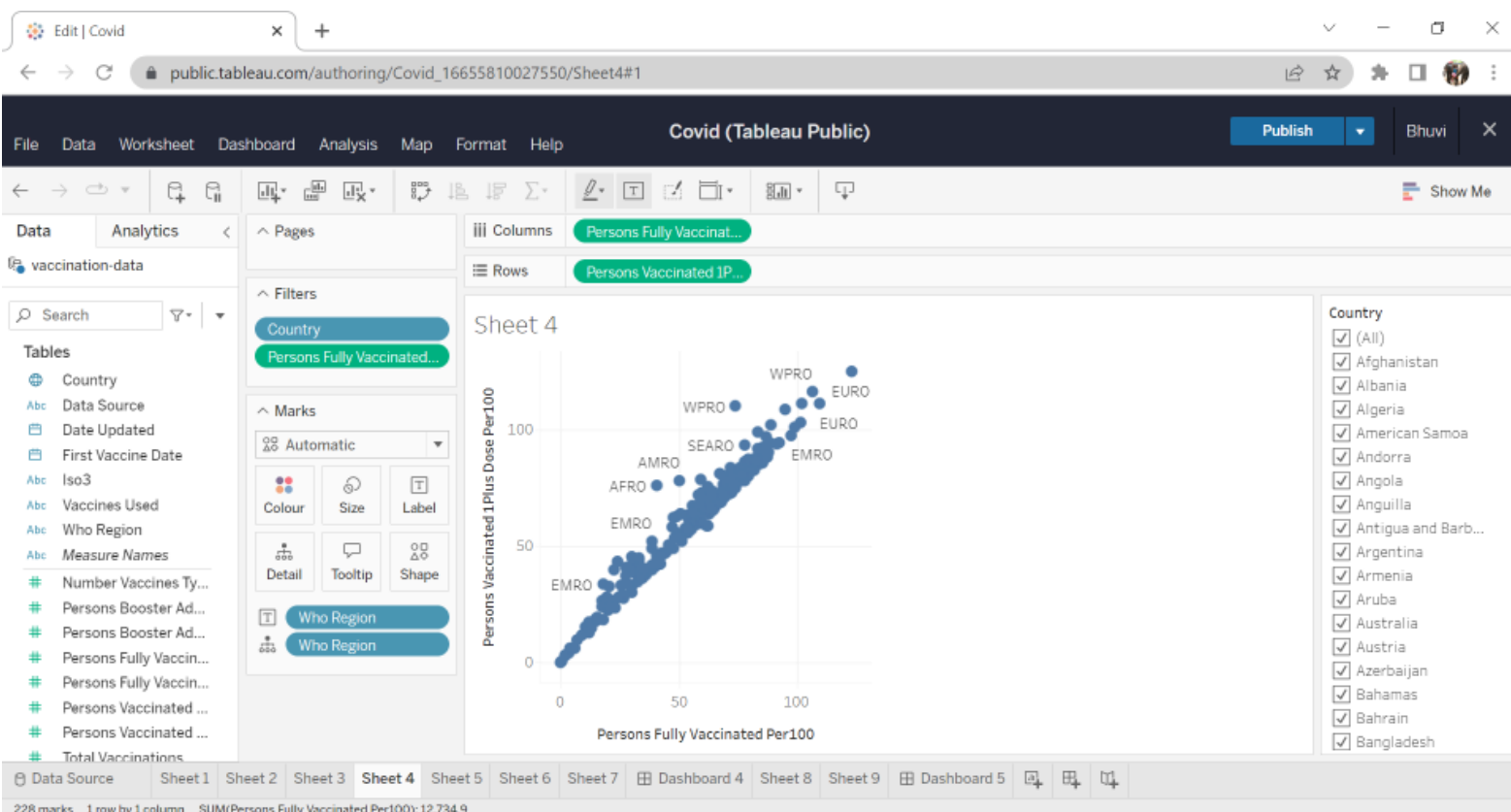
b) To understand if the covid vaccination scheme was successful in countries of Africa, percentage of total number of vaccinated people for each country in Africa is depicted using a bar graph. It can be concluded that the scheme was successful only in a few countries in Africa.



c)Percentage of vaccinated population is shown using a map & the percentage can be interpreted with the help of colour shades assigned depending on the percentage.



d) A scatter plot of percentage of people vaccinated with +1 dose & people fully vaccinated is plotted & segregated on the basis of region.



e) All the above sheets displayed in a single interactive dashboard. This gives us information about the total percentage of people vaccinated, percentage of people vaccinated with +1 dose & percentage of people fully vaccinated in each country. Filters are added for the necessary information to be extracted.

